

IAL Physics Unit 2 — Circuit Rules: Kirchhoff's Laws, Series & Parallel | Recall Pack

IAL Physics Unit 2 — Circuit Rules: Kirchhoff's Laws, Series & Parallel

The whole topic in one mental model. Two conservation laws run every circuit. **Charge is conserved** → Kirchhoff's First Law (KCL): at every junction $\sum I_{in} = \sum I_{out}$ — this holds in *any* circuit, series or parallel, always, because charge can never pile up or vanish at a point. **Energy is conserved** → Kirchhoff's Second Law (KVL): around *any* closed loop $\sum \varepsilon = \sum IR$ — the e.m.f. given to each coulomb equals the total p.d. it drops on that single trip round one loop. From these two laws everything else falls out: in **series**, current is the same and voltages add, so $R_T = \sum R_i$ (resistance *increases*); in **parallel**, voltage is the same and currents add, so $\frac{1}{R_T} = \sum \frac{1}{R_i}$ (resistance *decreases*, always below the smallest branch). For combination networks: collapse each parallel sub-circuit first, then add series values. The single golden rule for proofs (Alex, 52:53): *start from the quantity that adds up* — series → voltage adds; parallel → current adds.

1. Jargon glossary

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Kirchhoff's First Law (KCL)	At any meeting-point of wires, the current flowing in equals the current flowing out — none gets lost, none piles up.	A statement of conservation of charge : at any junction $\sum I_{in} = \sum I_{out}$. Holds at every junction of every circuit, at all times.	$\sum I_{in} = \sum I_{out}$; current in A	NOT “a rule only for parallel circuits”; NOT a rule about voltage; NOT optional — it always applies.	Alex: “the charges don’t go anywhere; they can’t disappear, can’t be created or destroyed.”	Phuc trap (topology misconception — #1 for this topic) : at 36:04-area discussion Phuc treated KCL as a series-vs-parallel rule. It is a charge-conservation rule that holds at every junction in every circuit. In series it is trivially true (I same everywhere).

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Kirchhoff's Second Law (KVL)	Going once round any closed loop, the energy a battery gives each coulomb is exactly the energy that coulomb drops across the components on that loop.	A statement of conservation of energy : around any closed loop $\sum \varepsilon = \sum IR$. The charge returns to its start with no net energy gain.	$\sum \varepsilon = \sum IR$; volts V	NOT a per-component rule — it is per <i>loop</i> ; you do NOT add both parallel branches in one loop.	Alex: 1 C gains ε J at the cell and drops IR_k J at each resistor on the loop.	Phuc trap (double-counting, 36:04) : in the J-K-L-M circuit ($\varepsilon = 9.0$ V, $V_K = 7.0$ V, parallel $V_L = V_M = 2.0$ V) Phuc summed $7 + 2 + 2 = 11$ V. A charge takes one branch: any single loop gives $7 + 2 = 9.0$ V = ε .
Junction (node)	A point where three or more wires meet, so the current can split or recombine.	A circuit node where branch currents merge/divide; KCL is applied here. Only appears where branches exist (parallel sections).	node; —	NOT present in a pure series circuit (one wire in, one wire out); NOT a place where charge accumulates.	Alex draws arrows into/out of each junction to keep I_{in}/I_{out} straight.	Common trap : in complex networks, mark current-direction arrows on the diagram first, then apply KCL at each junction independently.
Closed loop	Any complete path you can trace with your finger that starts and ends at the same point without lifting.	A closed conducting path round which KVL is written; $\sum \varepsilon = \sum IR$ for that loop.	loop; —	NOT the whole circuit necessarily — a network has several loops; KVL is written per loop.	A loop through K then L (not M) gives $V_K + V_L = 9.0$ V.	Common trap : pick one complete path only. Through a parallel pair, your loop uses <i>one</i> branch, never both.
Series connection	Components joined end-to-end in a single line, like beads on one string — one path only.	Components sharing a single current path, no branch junctions between them. Same current; p.d.s add.	$I_1 = I_2 = I$; $V_T = \sum V_i$	NOT parallel; current is NOT shared/split; voltage is NOT the same across each component.	Worksheet 1a: $30 + 20 + 20 \Omega$ in series $\Rightarrow R_T = 70 \Omega$.	Phuc trap (13:49) : Phuc reasoned "I is same in series, so V is same." Wrong — $V = IR$: same I but different R gives different V . Only current is uniform in series.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Parallel connection	Components connected side-by-side between the same two points — current has a choice of routes.	Branches sharing two common nodes; same p.d. across each; currents add at the junction.	$V_1 = V_2 = V$; $I_T = \sum I_i$	NOT series; voltage is NOT shared/divided; resistance does NOT add directly.	Worksheet 1b: $60\ \Omega \parallel 30\ \Omega \Rightarrow R_T = 20\ \Omega$.	Common trap: parallel total resistance is always less than the smallest branch . If you get a value larger than any branch, you added resistances directly by mistake.
Current I	The rate at which charge flows past a point — how many coulombs per second go by.	Rate of flow of charge: $I = \frac{\Delta Q}{\Delta t}$. Conserved at junctions (KCL).	I ; amperes A = C s ⁻¹	NOT energy; NOT the same as voltage; NOT consumed by resistors (charge flows through).	Charge is physical (electrons); it is conserved, never stored in a wire.	Phuc trap (27:10): Phuc said “current of A is 2.0 voltage” — confused a 2.0 V value with a current. Keep I (A) and V (V) labelled separately; corrected to $I = 0.40$ A.
Potential difference (p.d.) V	The energy each coulomb gives up between two points — the “drop” across a component.	Energy transferred from electrical to other forms per unit charge: $V = \frac{E}{Q}$. One volt \$= \$ one joule per coulomb.	V ; volts V = J C ⁻¹	NOT the same as current; NOT uniform in series; NOT “stored” — it is energy per charge transferred.	“Charlie the Coulomb” loses 60 J/C at R_1 , then 40 J/C at R_2 (100 V e.m.f.).	Phuc trap (8:53): Phuc wrote $V = E/t$. The correct definition is $V = E/Q$ (energy per charge , not per time).
Electromotive force (e.m.f.) ε	The “push” energy the source gives to every coulomb it sends round the loop.	Energy converted from chemical (etc.) to electrical form per unit charge by the source. In KVL it is the energy <i>input</i> term.	ε ; volts V = J C ⁻¹	NOT a force (despite the name); NOT a p.d. drop — it is the energy <i>supplied</i> .	In the J-K-L-M loop the source supplies $\varepsilon = 9.0$ V to each coulomb.	Common trap: e.m.f. is energy <i>given to</i> charges; p.d. is energy <i>taken from</i> them. Opposite energy directions, same units.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Resistance R	How hard a component fights the flow of charge — turning electrical energy into heat.	$R = \frac{V}{I}$. Property of a component; converts electrical energy to thermal energy.	R ; ohms $\Omega = \text{V A}^{-1}$	NOT $R = IV$; NOT a store of charge or electrons.	Resistors dissipate energy as heat; charge passes straight through unchanged.	Phuc trap (Trap 2): $R = IV$ is wrong — gives wrong units ($\$ \neq \$ \text{V} \cdot \text{A}$). Always $R = V/I$. Mnemonic “Very Itchy Rash”: $I = V/R$, $R = V/I$.
Total / equivalent resistance R_T	The single resistor that could replace the whole network and behave identically.	The combined resistance: series $R_T = \sum R_i$; parallel $\frac{1}{R_T} = \sum \frac{1}{R_i}$.	R_T ; ohms Ω	NOT $\sum R_i$ for parallel; NOT larger than the smallest branch in parallel.	Combination $80 \Omega + (20 \parallel 10) \Omega = 86.7 \Omega$.	Phuc trap (43:40): Phuc started the <i>parallel</i> proof from $R_1 + R_2$ (the series move) and got stuck in a fraction. Start parallel proofs from $I_T = I_1 + I_2$.
Combination network	A circuit mixing series and parallel parts — solve it in stages, not in one go.	A network reducible by repeatedly collapsing parallel sub-circuits, then adding series resistances.	mixed; Ω	NOT solvable by a single formula applied to the whole thing.	OneNote tile 3: 7.0Ω series with $4.0 \parallel 12 \Omega = 3.0 \Omega \Rightarrow R_T = 10.0 \Omega$.	Phuc trap (55:22-56:44): Phuc applied one parallel formula to the whole combination. Two-step method: solve the parallel section first, <i>then</i> add the series resistor.

2. Core laws — formula / case table

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Kirchhoff 1 (KCL)	$\sum I_{in} = \sum I_{out}$	I_{in} = currents entering a junction; I_{out} = currents leaving it	A	Any junction in any circuit; finding a missing branch current.	$I_{missing} = \sum I_{in} - \sum (\text{other } I_{out})$	Topology-independent — holds in series too (trivially). Not a “parallel-only” rule.	At a node: 3.0 A in = I_L + 2.0 A out $\Rightarrow I_L = 1.0$ A (J-K-L-M).
Kirchhoff 2 (KVL)	$\sum \varepsilon = \sum IR$	ε = source e.m.f.s on the loop; IR = p.d. drops on the loop	V	Any closed loop; relating e.m.f. to the p.d. drops on one path.	$\varepsilon = V_1 + V_2 + \dots$	One loop = one path. Do not add both parallel branches in a single loop.	Loop K $\rightarrow$ L: $V_K + V_L = 7.0 + 2.0 = 9.0$ V = ε (NOT 7 + 2 + 2).
Definition of current	$I = \frac{\Delta Q}{\Delta t}$	ΔQ = charge passing; Δt = time taken	I in A; Q in C; t in s	Linking charge flow to current; conservation arguments.	$\Delta Q = I \Delta t$	Charge is conserved (KCL); it is not used up at resistors — only energy is.	6.0 C in 2.0 s $\Rightarrow I = 3.0$ A.
Definition of p.d. / e.m.f.	$V = \frac{E}{Q}$	E = energy transferred; Q = charge	V = J C ⁻¹	Defining p.d. (energy out) or e.m.f. (energy in); “Charlie the Coulomb” reasoning.	$E = VQ$	Phuc’s 8:53 slip: it is energy per charge E/Q , NOT per time E/t .	100 J given to 1 C at the cell $\Rightarrow \varepsilon = 100$ V.
Ohm’s law (definition of R)	$R = \frac{V}{I}$, i.e. $V = IR$	V = p.d. across; I = current through; R = resistance	V, A, Ω	Finding any one of V, I, R from the other two; per-component working.	$I = \frac{V}{R}$; $V = IR$	Phuc trap (Trap 2): never $R = IV$ (wrong units). “Very Itchy Rash” $\Rightarrow I = V/R$, $R = V/I$.	$V = 2.4$ V across $R = 6.0 \Omega \Rightarrow I = 0.40$ A.

3. Series & parallel rules — one-glance comparison

Quantity / case	Series rule	Parallel rule	Symbol meanings	When to use	Trap	Mini worked example
Current	Same everywhere: $I_1 = I_2 = I$	Splits: $I_T = I_1 + I_2 + \dots$	I = current; subscripts = components/branches	Deciding what is uniform before substituting	In series, current — not voltage — is the same. (Phuc 13:49)	Series: ammeters read identical I . Parallel: $0.80 + 0.40 = 1.20$ A.
Potential difference	Shared (adds): $V_T = V_1 + V_2 + \dots$	Same across each branch: $V_1 = V_2 = V$	V = p.d.	Deciding what is uniform; KVL set-up	In series V differs per component ($V = IR$, same I , different R). (Phuc 13:49)	Series: $9.6 + 2.4 = 12.0$ V. Parallel: both branches at 2.4 V.
Total resistance	$R_T = R_1 + R_2 + \dots$	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	R_T = equivalent resistance	Collapsing a sub-circuit to one resistor	Parallel: take the reciprocal at the end. Never add parallel R directly. (Trap 4)	Series: $8.0 + 2.0 = 10.0 \Omega$. Parallel: $\frac{1}{3} + \frac{1}{6} = \frac{1}{2} \Rightarrow R_T = 2.0 \Omega$.
Effect of adding a resistor	R_T increases	R_T decreases (always < smallest branch)	—	Sanity-checking your answer	Parallel result larger than a branch $\Rightarrow$ you added directly. (Trap 3)	Adding a 4th parallel branch lowers R_T further.
Two-resistor parallel shortcut	(n/a)	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ (product over sum)	R_1, R_2 = the two branch resistances	Exactly two parallel resistors	Product-over-sum works for two only; use $1/R_T$ sum for three+.	$\frac{3.0 \times 6.0}{3.0 + 6.0} = \frac{18}{9} = 2.0 \Omega$.
Proof starting point	Start from $V_T = V_1 + V_2$ (KVL, voltage adds)	Start from $I_T = I_1 + I_2$ (KCL, current adds)	—	Deriving R_T from scratch (state-the-proof questions)	Phuc 43:40 / Alex 52:53: start from the quantity that <i>adds</i> in that circuit type.	Parallel from $I_T = I_1 + I_2 \rightarrow$ divide by $V \rightarrow \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$.

4. Derivations to memorise (state-the-proof questions)

Proof	Step 1 (starting law)	Step 2 (Ohm's law)	Step 3 (uniform quantity)	Step 4 (cancel)	Result	Trap
Series R_T	KVL: $V_T = V_1 + V_2$	$I_T R_T = I_1 R_1 + I_2 R_2$	series current rule $I_T = I_1 = I_2 = I$	divide by I	$R_T = R_1 + R_2$	Must start from <i>voltage adds</i> ; show the cancel- I step for full marks.
Parallel R_T	KCL: $I_T = I_1 + I_2$	$\frac{V_T}{R_T} = \frac{V_1}{R_1} + \frac{V_2}{R_2}$	parallel voltage rule $V_T = V_1 = V_2 = V$	divide by V	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	Phuc 43:40: do NOT start from $R_1 + R_2$. Start from $I_T = I_1 + I_2$, write Ohm's law as $I = V/R$.

5. Combination-network method (two-step golden rule)

- **Step 1 — collapse the parallel section first.** Compute the parallel sub-circuit's R_p using $\frac{1}{R_p} = \sum \frac{1}{R_i}$ (or product-over-sum for two). *Never* apply a single formula to the whole network at once (Phuc's 55:22 error).
- **Step 2 — add series resistances.** $R_T = R_{series} + R_p$.
- **Step 3 — total current** from $I_T = \varepsilon / R_T$.
- **Step 4 — series p.d.** across each series element: $V = I_T R$ (same current).
- **Step 5 — parallel-section p.d.** by KVL: $V_p = \varepsilon - V_{series}$.
- **Step 6 — branch currents** in the parallel section: $I_i = V_p / R_i$.
- **Always check both laws:** KCL ($\sum I_{branch} = I_T$) and KVL ($\sum V = \varepsilon$).

Worked instance (Example 1, Figure 4): 12.0 V ; $R_1 = 8.0 \Omega$ series with $R_2 = 3.0 \Omega \parallel R_3 = 6.0 \Omega$. $R_{23} = 2.0 \Omega \Rightarrow R_T = 10.0 \Omega \Rightarrow I_T = 1.2 \text{ A}$. $V_1 = 1.2 \times 8.0 = 9.6 \text{ V}$; $V_{23} = 12.0 - 9.6 = 2.4 \text{ V}$; $I_2 = 2.4/3.0 = 0.80 \text{ A}$, $I_3 = 2.4/6.0 = 0.40 \text{ A}$. Check KCL: $0.80 + 0.40 = 1.20 \text{ A} \checkmark$; KVL: $9.6 + 2.4 = 12.0 \text{ V} \checkmark$.

6. Exam traps & mark-scheme keywords

Trap	Why it bites	The fix	Evidence
KCL is topology-independent	Treated as a "parallel-only" rule	KCL = conservation of charge; holds at every junction, every circuit. In series it is trivially satisfied.	#1 misconception for this topic; Phuc

Trap	Why it bites	The fix	Evidence
$R = IV$	Wrong units ($\Omega \neq V \cdot A$), wrong answers	Always $R = V/I$. “Very Itchy Rash”.	Trap 2; lesson emphasis
Parallel $R_T >$ smallest branch	Added resistances directly	R_T in parallel is always below the smallest branch; reciprocal-sum then invert.	Trap 3
Adding parallel R directly	$R_T \neq R_1 + R_2$ for parallel	Use $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$; remember the final reciprocal.	Trap 4
Unit prefixes	mA / μ A / k Ω mixed up	1 mA = 10^{-3} A; 1 μ A = 10^{-6} A; 1 k Ω = $10^3 \Omega$. Convert to base SI before substituting.	Trap 5
p.d. same in series	“Same $I \Rightarrow$ same V ”	$V = IR$: same I , different $R \Rightarrow$ different V . Only I is uniform in series.	Trap 6; Phuc 13:49
“Resistors store electrons”	Confusing charge with energy	Resistors resist flow and dissipate energy as heat; charge passes through, none accumulates.	Trap 7
Double-counting parallel branches in KVL	Summed both branches in one loop	One loop uses one branch: $\varepsilon = V_{series} + V_{one\ branch}$.	Trap 8; Phuc 36:04
Combination: not solving parallel first	One formula on the whole network	Collapse the parallel sub-circuit, then add series.	Trap 9; Phuc 55:22
Not stating the law	“State and explain” loses marks	(1) name the law, (2) state it in words with the conservation principle, (3) write the equation, (4) apply.	Trap 10

Mark-scheme keywords - **KCL**: “consequence of conservation of charge”; “sum of currents into a junction equals the sum out”. - **KVL**: “consequence of conservation of energy”; “sum of e.m.f.s equals the sum of p.d.s around a loop”. - **Series derivation**: must show $V_T = V_1 + V_2$, then $IR_T = IR_1 + IR_2$, then cancel I . - **Parallel derivation**: must show $I_T = I_1 + I_2$, then $V/R_T = V/R_1 + V/R_2$, then cancel V . - Always show working — “no ecf if no working” is standard.